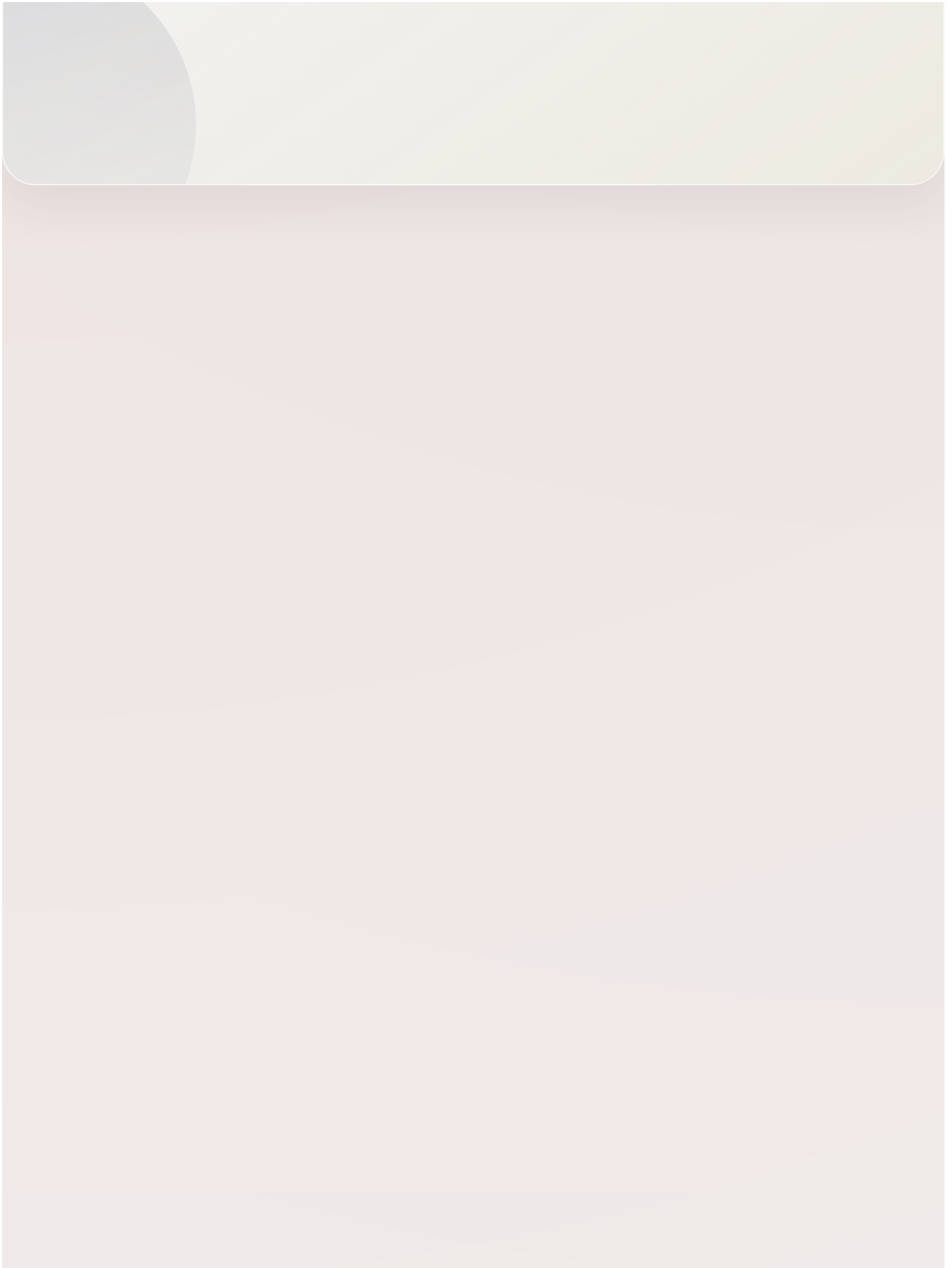


MYAVANA[®]

TECHNICAL DOCUMENTATION



SYSTEM SNAPSHOT

PLATFORM SUMMARY

PRODUCTION READY

The MYAVANA Hair Journey platform is implemented as a feature-rich WordPress plugin that turns a standard WP site into a logged-in hair wellness application. The plugin owns the customer-facing product surface, authentication modal, onboarding, journey logging, goal/routine management, unified profile, AI features, community feed, and internal intelligence dashboards.

6

Core primary pages in the user-facing experience: Home, My Timeline, Profile, Community, Goals, Routines.

20+

Shortcodes registered to map feature surfaces into pages and reusable containers.

3

Major storage models: WordPress posts/meta, user meta, and custom Myavana tables.

4

Primary interaction pillars: Journey, AI, Community, Gamification.

Core architectural decision: the plugin deliberately mixes modern application behavior with WordPress-native primitives. Journey entries are persisted as a custom post type, long-lived personalized state sits in user meta, and high-volume or relational features such as social activity and gamification are stored in dedicated custom tables.

USER SURFACE

SITE PAGES AND USER CAPABILITIES

Home

Marketing and launch surface driven by premium MYAVANA styling and entry points into authentication and analysis.

- Guest conversion surface
- Modal-driven sign in / sign up
- Launch into onboarding and first-entry experience
- Supports domain-based production entry and Google auth

My Timeline / Hair Journey

Primary logged-in workspace for entries, goals, routines, analytics summaries, and AI-informed progress review.

- Timeline, calendar, and list views
- Create/edit offcanvas forms for entries, goals, and routines
- Media-backed entry logging
- Quick launch for first entry after onboarding

Profile

Community

Unified personal dashboard combining identity, community stats, hair profile data, and AI/gamification summaries.

- Hair profile and community profile integration
- Completion score and suggested next actions
- Snapshot history and insight summaries
- Owner-aware editing behavior

Social layer for sharing entries, discovering other users, following creators, reacting to posts, and participating in circles.

- Discover and following modes
- Profile widget and user stats
- Search, media filtering, and circle filters
- Entry-to-community publishing

Goals

Goal management layer backed by structured user meta and milestone-aware reward handling.

- Structured goal objects
- Progress history and milestone tracking
- Target dates, categories, priorities, and metrics
- Reward hooks at 25/50/75/100 percent

Routines

Routine planning and completion tracking with due-today logic, completion history, and performance summaries.

- Routine frequency normalization
- Due-date calculation and completion map
- Step-based routines
- Bridge into timeline and community library

BOOTSTRAP

PLUGIN ENTRY ARCHITECTURE

The plugin bootstraps from `myavana-hair-journey.php`. This file registers the plugin, loads supporting includes/actions/templates, instantiates the main application class, and coordinates page creation, script loading, navigation, protection rules, and legal page seeding.

Plugin file

| loads includes/ (auth, AI, data, analytics, errors, social, gamification)

| loads actions/ (entries, goals/routines, community, insights, dashboard handlers)

| loads templates/ page shortcode implementations

| instantiates Myavana_Hair_Journey

| activates default pages on plugin activation

| registers shortcodes on init

| enqueues global + page-specific assets

| renders navigation, preloader, mobile tabs

| enforces access and logout behavior

BOOTSTRAP CONCERN	IMPLEMENTATION	OBSERVED BEHAVIOR
Main runtime	<code>class Myavana_Hair_Journey</code>	Central orchestrator for hooks, activation, assets, navigation, and protected route behavior.
Activation	<code>register_activation_hook(..., activate)</code>	Creates baseline profile table and seeds default pages containing shortcodes.
Page seeding	<code>get_default_pages()</code>	Creates Login, Register, Profile, Timeline, Goals, Routines, Add Entry, Try-On, Hair Insights, Privacy, and Terms pages.
Legal page safety	<code>maybe_seed_legal_pages()</code>	Ensures privacy and terms pages exist for active admin sessions even if activation-time creation was missed.
Protected routes	<code>template_redirect</code> enforcement	Restricts authenticated app surfaces to signed-in users while preserving a guest-facing homepage/auth path.
Logout behavior	<code>logout_redirect</code> filter	Redirects users back to the site homepage instead of a stock WP login endpoint.

ROUTING

SHORTCODE AND PAGE COMPOSITION MODEL

The product is page-composed through shortcodes registered by `includes/shortcodes.php`. This creates a WordPress-native routing layer where editable pages can host complex application modules.

SHORTCODE	PRIMARY SURFACE	FUNCTION TARGET
[myavana_luxury_home]	Home	myavana_luxury_home_shortcode()
[myavana_hair-journey-page]	My Timeline / Hair Journey	myavana_hair_journey_page_shortcode()
[myavana_unified_profile]	Profile	myavana_unified_profile_shortcode()
[myavana_community_feed]	Community	myavana_community_feed_shortcode()
[myavana_goals_page]	Goals	myavana_goals_page_shortcode()
[myavana_routines_page]	Routines	myavana_routines_page_shortcode()
[myavana_login]	Legacy login page	myavana_login_shortcode()
[myavana_register]	Legacy registration page	myavana_register_shortcode()
[myavana_privacy_policy]	Privacy page	myavana_privacy_policy_shortcode()
[myavana_terms]	Terms page	myavana_terms_shortcode()

Operational advantage: this shortcode model gives non-developers page-level control inside WordPress while allowing developers to keep feature implementations modular in PHP templates and JS/CSS bundles.

IDENTITY

AUTHENTICATION AND ONBOARDING

Authentication Runtime

`includes/myavana-auth-system.php` encapsulates the auth stack.

- Admin settings page for modal behavior, onboarding, auto-approval, and Google auth
- Frontend auth modal injected via `wp_footer`
- AJAX handlers for standard auth, Google auth, password reset, onboarding save/step/skip/reset, and onboarding triggers
- Rewrite-backed password reset route handling
- Registration hooks and login hooks for onboarding state management

Google Auth Model

- Configured as a web client ID in plugin settings
- Uses Google Sign-In through frontend JavaScript origin validation, not a redirect URI callback flow
- Server endpoint: `myavana_google_auth`
- Onboarding can trigger after Google registration/sign-in for pending users
- Designed to keep users in the MYAVANA modal flow instead of WP login screens

1 Guest lands on Home or other enabled surface and the auth modal system is eligible to render.

2 User signs in via email/password or Google Sign-In from the branded modal.

3 Server validates user state, creates or locates the account, and returns onboarding trigger state.

4 Onboarding persists hair profile and preference data through AJAX step handlers.

5 User is redirected into the app surface with first-entry launch support and homepage-safe logout behavior.

Default behavior as audited: onboarding is enabled by default when no explicit option override exists, and Google auth has been integrated into the same onboarding-aware auth lifecycle as native registration.

JOURNEY CORE

HAIR JOURNEY PAGE ARCHITECTURE

The My Timeline page is assembled by `templates/pages/hair-journey.php`. It is a logged-in-only workspace built from shared data retrieved once through `Myavana_Data_Manager::get_journey_data()`, then rendered through smaller partials.

PARTIAL	PURPOSE	NOTES
header-and-sidebar.php	Navigation, sidebar summaries, global controls	Works with shared analytics, profile, and gamification data.
timeline-area.php	Main journey content wrapper	Hosts timeline, calendar, and list view containers plus actions.
view-offcanvas.php	Detail viewer offcanvas	Supports entry/goal/routine inspection from timeline interactions.
create-offcanvas.php	Create/edit forms	Handles entry, goal, and routine form UX with AJAX submission.

View Modes

- **Timeline:** editorial card-based chronology with entry, goal, and routine blocks
- **Calendar:** date-centric review of activity
- **List:** denser chronological scanning view

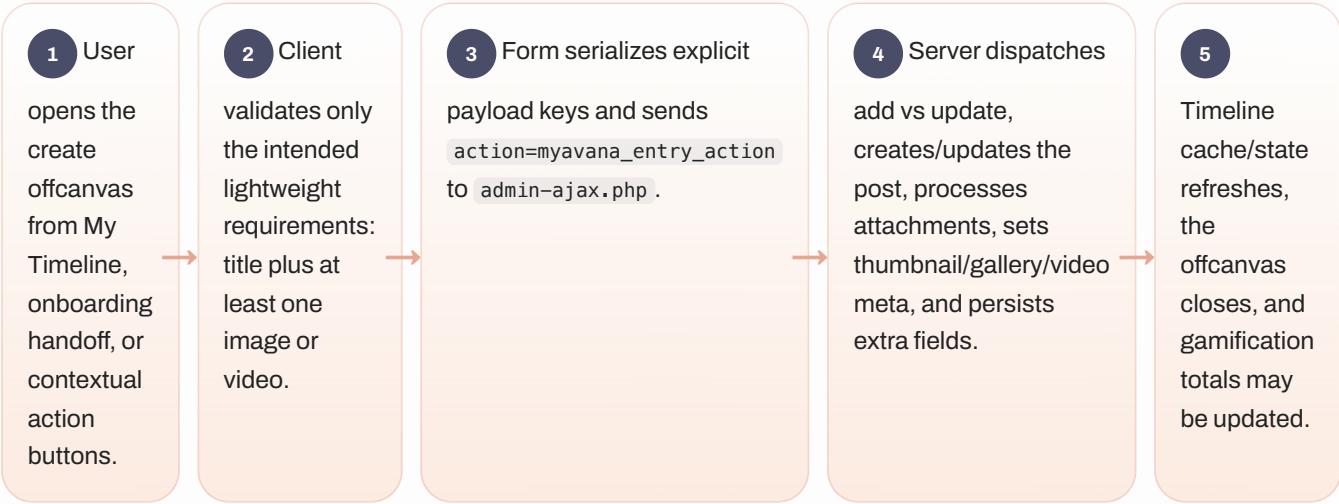
Launch Enhancements

- Welcome banner and first-entry auto-open via query flags
- Preloaded `window.myavanaEntryCache` to reduce follow-up fetches
- Analysis limit awareness via shared data manager payload
- Journey launch state persisted in user meta for onboarding handoff

ENTRY FLOW

HAIR JOURNEY ENTRY LIFECYCLE

Entries are the central unit of the platform. They are stored as WordPress posts of type `hair_journey_entry` and enriched through post meta plus attachment relationships for images and video.



LAYER	FILE(S)	RESPONSIBILITY
Form UX	templates/pages/partials/create-offcanvas.php	Multi-tab entry form, previews, validation, explicit payload assembly, and submit interception.
Dispatch	actions/hair-entries.php	myavana_handle_entry_action() routes add vs update by presence of entry_id.
Create / update persistence	templates/hair-diary-timeline-shortcode.php	Holds the active myavana_add_entry() and myavana_update_entry() implementations used by the dispatcher.
Modular timeline forms	assets/js/timeline/timeline-forms-new.js	Secondary timeline module that cooperates with HJN-owned forms and avoids duplicate submit binding.

Current validated create rule: the front end and server were aligned so that manual entry creation requires a title and at least one uploaded image or video. Rating is optional and no longer blocks creation.

STRUCTURED PLANNING

GOALS AND ROUTINES SUBSYSTEMS

Goals	Routes
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- Stored in user meta key `myavana_hair_goals_structured`
- Managed through `actions/goal-routine-handlers.php`
- Fields include title, description, category, priority, start/target dates, progress, measurement values, milestones, and notes
- Progress milestone rewards are issued at 25/50/75/100 percent thresholds
- Goal pages normalize categories, metrics, and status for a premium management dashboard

- Stored in user meta key `myavana_current_routine`
- Completion history stored in `myavana_routine_completions`
- Routines page computes normalized frequency classes such as daily, twice weekly, bi-weekly, monthly, and as needed
- Supports reminder-day aware due-today logic and rolling completion analysis
- Timeline and routines page share the same backing state, allowing direct cross-surface consistency

User Meta Backing

```
myavana_hair_goals_structured    -> array of structured goal records
myavana_current_routine         -> array of routine records
myavana_routine_completions     -> map: YYYY-MM-DD => [routine_ids]
```

Operational Pattern

```
form submit -> AJAX handler -> sanitize / normalize -> update user meta -> optional reward check
```

SOCIAL LAYER

COMMUNITY ARCHITECTURE

Community capabilities are implemented as a relational subsystem in `includes/social-features.php`. Unlike goals and routines, this layer is not stored in generic user meta because it requires multi-entity queries, counters, follow relationships, bookmarks, and post-level interactions at scale.

TABLE	PURPOSE	KEY RELATIONSHIPS
<code>myavana_community_posts</code>	Community feed content	User-authored posts, source entry linkage, counts, media, privacy
<code>myavana_post_likes</code>	Like state	Unique (<code>post_id</code> , <code>user_id</code>)
<code>myavana_post_comments</code>	Comments and replies	Post, user, optional parent comment
<code>myavana_user_followers</code>	Follow graph	Follower to following user relationship
<code>myavana_community_challenges</code>	Challenge catalog	Date ranges, rules, participation counts
<code>myavana_challenge_participants</code>	User challenge state	Challenge/user uniqueness, progress data
<code>myavana_shared_entries</code>	Journey-to-community mapping	One shared community post per source entry
<code>myavana_shared_routines</code>	Routine library	Shareable routines with metadata and engagement signals
<code>myavana_post_bookmarks</code>	Saved posts	Collection-aware bookmarking
<code>myavana_notifications</code>	Notification queue	User-scoped, read-tracked notification records

Community page behavior: the visible feed couples database-backed social objects with profile context, discovery filters, search, circles, following mode, and direct entry-sharing actions. This gives the product a social graph without forcing the journey model itself into a social-table-first design.

REWARDS

GAMIFICATION AND ENGAGEMENT ENGINE

`includes/gamification.php` provides a dedicated engagement subsystem with its own schema versioning, runtime migration checks, reward lookups, stats synchronization, badges, points history, insights, and AI tip storage.

Progress Tracking

- Daily check-ins
- Current and longest streak
- Total entries and AI analyses
- User level and XP progress

Reward Objects

- Default seeded badges
- Points history by event key
- Unlocked insights
- Recent rewards / recent badges

Operational Hooks

- Goal progress milestone rewards
- Entry media reward paths
- Quest/challenge scaffolding
- User summary retrieval for dashboards

GAMIFICATION TABLE	ROLE
myavana_daily_checkins	Calendar-based engagement and streak events
myavana_user_stats	User aggregate points, streaks, totals, level
myavana_badges	Badge catalog definitions
myavana_user_badges	User-earned badge state
myavana_points_history	Audit log of reward changes by event key
myavana_insights	Unlockable insight catalog
myavana_user_insights	User insight unlock record
myavana_ai_tips	Served AI tips and click tracking

AI LAYER

AI, RECOMMENDATIONS, AND GENERATED MEDIA

Gemini-Based Services

- `Myavana_AI` is the core AI service wrapper
- Image preview model: `gemini-2.0-flash-preview-image-generation`
- Text/vision analysis model: `gemini-2.0-flash`
- API keys are read from WordPress options, not hardcoded constants
- Daily per-user usage caps are enforced via user meta counters

AI Capabilities in Product

- Hairstyle preview generation from uploaded images
- Comprehensive hair analysis from photo input
- Auto-generated journey entries from AI analysis paths
- Insight summaries in profile and analytics surfaces
- AI-assisted product/style metadata extraction

Implementation note: generated AI results do not live only as transient UI artifacts. The system can persist relevant analysis output back into post meta, user profile snapshots, and journey history, which is critical for longitudinal progress views and recommendation continuity.

STORAGE MODEL

DATA PERSISTENCE STRATEGY

STORAGE TYPE	USED FOR	EXAMPLES
Custom post type + post meta	Hair journey entries and attached media	<code>hair_journey_entry</code> , <code>health_rating</code> , <code>mood_demeanor</code> , <code>_entry_gallery</code> , <code>_entry_videos</code>
User meta	User-scoped personalized state	Goals, routines, onboarding progress, typeform data, first-entry prefill, analysis usage, routine completions
Custom relational tables	Analytics, profiles, social features, gamification	<code>myavana_profiles</code> , <code>myavana_site_analytics</code> , community tables, gamification tables

Profile/Analytics Base Tables

- `myavana_profiles`
- `myavana_conversations`
- `myavana_site_analytics`

Important User Meta Keys

- `myavana_hair_goals_structured`
- `myavana_current_routine`
- `myavana_routine_completions`

- myavana_hair_analysis_history
- myavana_first_entry_prefill

Storage Heuristic

If the entity is editorial and timeline-visible -> custom post type + post meta
If the entity is private, per-user, and array-friendly -> user meta
If the entity requires joins, counters, or social graph behavior -> custom table

ASSETS

FRONTEND BUNDLE AND UI MODULE STRATEGY

The frontend is heavily asset-driven. The main plugin class enqueues a shared visual foundation, then conditionally loads page- specific bundles such as the home page, global navbar, timeline modules, Select2, FilePond, Splide, and AI modal assets.

BUNDLE GROUP	REPRESENTATIVE ASSETS	ROLE
Brand foundation	assets/css/myavana-styles.css , global fonts	Typography, tokens, visual baseline
Navigation	global-navbar.css , global-navbar.js , mobile tabs	Cross-page navigation and mobile app-shell feel
Auth	myavana-auth-system.js , auth modal template	Modal auth, Google sign-in, onboarding triggers
Timeline app	assets/js/timeline/* , timeline partial CSS	State, offcanvas, navigation, list/timeline view, filters, comparison
Media UX	FilePond, image preview, Select2	Uploads, rich form controls, multi-select product behavior
Discovery/social	social-feed.js , share-to-community.js	Community feed behaviors and sharing interactions

Design implementation pattern: the plugin keeps page-level visual language intentionally branded and highly custom. Rather than leaning on generic admin-looking forms, it uses editorial cards, glass surfaces, strong typography, and mobile-specific layout overrides to preserve an app-like feel inside WordPress.

ANALYTICS AND ADMIN

OPERATIONAL INTELLIGENCE LAYER

Frontend Analytics Capture

- AJAX tracking endpoint for event batches
- Performance metrics ingestion endpoint
- Session, path, title, device, viewport, and metadata capture
- WebSocket auth token endpoint for realtime integrations

Admin Intelligence

- Dedicated Myavana admin menu and dashboard redirect
- Overview, Executive, Growth, Community, Retention, AI Lab, Operations, and Gamification intelligence pages
- Settings page for xAI, OpenAI, and Gemini credentials

- Component lazy-load endpoint for modular rendering

- Chart.js-backed dashboard visualizations
- Role-filtered, date-filtered reporting helpers

SECURITY

SECURITY AND RELIABILITY PATTERNS

CONCERN	OBSERVED PATTERN	WHY IT MATTERS
AJAX authorization	Nonce verification is used broadly across auth, journey, goals, routines, profile, and admin handlers.	Prevents cross-site request forgery on mutation endpoints.
Input sanitation	Use of <code>sanitize_text_field</code> , <code>sanitize_textarea_field</code> , integer bounds, and structured normalization.	Reduces malformed payloads and unsafe persistence.
User ownership checks	Entry update handlers verify post authorship; goal/routine handlers read from current user meta only.	Prevents cross-user modification of personalized data.
Route protection	Logged-in-only app pages return sign-in prompts or are protected during redirect handling.	Protects personal journey and community management surfaces.
Schema safety	Custom tables are created through <code>dbDelta</code> ; some systems perform runtime schema checks and repairs.	Mitigates production drift across environments and plugin revisions.

Important engineering note: the codebase currently contains some legacy overlap, especially around older diary/timeline handlers and newer modular handlers. The active runtime works, but long-term maintenance will benefit from consolidating duplicated entry-related endpoints into a single canonical implementation.

PRODUCTION

DEPLOYMENT AND CONFIGURATION CHECKLIST

AREA	REQUIRED STATE	NOTES
Plugin version	<code>2.6.5</code>	Current audited build state.
Domain	Connected and serving the production WordPress instance	App shell and auth rely on correct origin behavior.
Google auth	Enabled with correct Web Client ID and JavaScript origin	Redirect URI is not needed for the current implementation.
API credentials	Gemini, OpenAI, and/or xAI keys stored in plugin settings as needed	AI features and admin intelligence depend on this configuration.
Legal pages	<code>/privacy</code> and <code>/terms</code> exist	Plugin can seed them automatically for admins.

AREA	REQUIRED STATE	NOTES
Database schema	Custom profile, social, gamification, analytics tables available	Runtime migrations help, but production should still be validated once.
Media uploads	WP uploads writable and attachment processing functioning	Entry creation now requires at least one uploaded image or video.

Recommended launch smoke test: guest landing page, email signup, Google signup, onboarding completion, first entry creation with media, goal creation, routine creation, timeline render on mobile, profile render, community share, logout redirect, and AI modal invocation.

FILE MAP

CORE SOURCE LOCATIONS

PATH	AREA	ROLE
myavana-hair-journey.php	Bootstrap	Main plugin entry, activation, page seeding, asset orchestration
includes/shortcodes.php	Routing	Shortcode registry and dispatch surface
includes/myavana-auth-system.php	Identity	Auth modal, Google auth, onboarding, password reset
includes/class-myavana-data-manager.php	Data aggregation	Shared journey data retrieval and caching
includes/myavana_database_setup.php	Schema	Profiles, conversations, analytics table creation
includes/social-features.php	Community	Social schema and social AJAX features
includes/gamification.php	Rewards	Points, streaks, badges, insights, summary engine
includes/ai-integration.php	AI	Gemini-based hair analysis and preview generation
actions/hair-entries.php	Journey dispatch	Entry action routing and related AJAX handlers
actions/goal-routine-handlers.php	Goals/Routines	CRUD, progress history, rewards integration
templates/pages/hair-journey.php	Timeline page	Main logged-in journey composition layer
templates/pages/partials/create-offcanvas.php	Entry UX	Create/edit forms and HJN client runtime
templates/hair-diary-timeline-shortcode.php	Entry persistence	Active add/update entry implementations and media handling
templates/pages/goals.php	Goals page	Goal dashboard and management rendering
templates/pages/routines.php	Routines page	Routine dashboard and completion analysis
templates/pages/community/community-feed.php	Community page	Social discovery and feed UI
templates/pages/unified-profile.php	Profile page	Hair journey + community + rewards profile surface

End of document. Generated from a direct source audit of the MYAVANA Hair Journey plugin and site implementation.